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VISUAL SYSTEMS AND STATISTICAL METHODS FOR NON-DESTRUCTIVELY EVALUATING COMPONENT PARTS

Abstract

A system and a method include a light emitter configured to emit light onto or into a part to illuminate the part. A detector is configured to acquire one or more digital images of the part as illuminated by the light. A control unit is in communication with the detector. The control unit is configured to receive the one or more digital images of the part as illuminated by the light from the detector, and automatically analyze the one or more digital images of the part (to normalize out, such as by removing, known geometric features) as illuminated by the light to one or both of detect one or more defects of the part, or determine a porosity of the part.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] Examples of the present disclosure generally relate to systems and methods for non-destructively inspecting component parts, such as translucent composite panels, beams, or other such structures.

BACKGROUND OF THE DISCLOSURE

[0002] During a manufacturing process, various structures can be inspected to ensure structural integrity. For example, an individual can visually inspect a structure for potential defects.

[0003] Visual inspection of component parts, such as translucent composite parts, can be subjective and imprecise. As such, strict guidelines for colorblindness and visual acuity may be required for individuals performing such inspections. As can be appreciated, however, such processes can be prone to human error.

SUMMARY OF THE DISCLOSURE

[0004] A need exists for an improved system and method for inspecting component parts to ensure structural integrity. Further, a need exists for an efficient and effective system and method for visually inspecting component parts without the subjectivity of the human eye.

[0005] With those needs in mind, certain examples of the present disclosure provide a system including a light emitter configured to emit light onto or into a part to illuminate the part. A detector is configured to acquire one or more digital images of the part as illuminated by the light. A control unit is in communication with the detector. The control unit is configured to receive the one or more digital images of the part as illuminated by the light from the detector, and automatically analyze the one or more digital images of the part as illuminated by the light to detect one or more defects of the part, and/or determine a porosity of the part. In at least one example, the digital images of the part are analyzed to normalize out, such as by removing, certain known geometric features.

[0006] In at least one example, the detector includes a digital camera. The light emitter can be separated from the part.

[0007] In at least one example, a user interface is in communication with the control unit. The user interface includes a display. The control unit is configured to show information regarding the part on the display.

[0008] In at least one example, the control unit is configured to automatically analyze the one or more digital images by determining a mean and a standard deviation of one or more portions of the one or more digital images.

[0009] In at least one example, the control unit is configured to automatically analyze the one or more digital images by determining a signal-to-noise ratio (SNR) of one or more portions of the one of more digital images.

[0010] In at least one example, the control unit is configured to analyze the one or more digital images to generate a histogram in relation to one or more portions of the one or more digital images, and determine a mean and a standard deviation from the histogram.

[0011] The control unit can be an artificial intelligence or machine learning system.

[0012] Certain examples of the present disclosure provide a method comprising: emitting, by the light emitter, the light onto or into the part to diffusively illuminate the part; acquiring, by the detector, the one or more digital images of the part as illuminated by the light; receiving, by the control unit, the one or more digital images of the part as illuminated by the light from the detector; and automatically analyzing, by the control unit, the one or more digital images of the part as illuminated by the light, wherein said automatically analyzing comprises one or both of detecting the one or more defects of the part, or determining the porosity of the part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. **1** illustrates a block diagram of a system, according to an example of the present disclosure.

[0014] FIG. **2** illustrates a flow chart of a qualitative assessment method, according to an example of the present disclosure.

[0015] FIG. **3** illustrates a flow chart of a quantitative assessment method, according to an example of the present disclosure.

[0016] FIG. **4** illustrates a front view of a digital image, according to an example of the present disclosure.

[0017] FIG. **5** illustrates a histogram, according to an example of the present disclosure.

[0018] FIG. **6** illustrates a flow chart of a method of detecting a defect in a digital image of a part, according to an example of the present disclosure.

[0019] FIG. **7** illustrates a flow chart of a method of detecting a porosity of a part, according to an example of the present disclosure.

[0020] FIG. **8** illustrates a schematic block diagram of a control unit, according to an example of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0021] The foregoing summary, as well as the following detailed description of certain examples will be better understood when read in conjunction with the appended drawings. As used herein, an element or step recited in the singular and preceded by the word “a” or “an” should be understood as not necessarily excluding the plural of the elements or steps. Further, references to “one example” are not intended to be interpreted as excluding the existence of additional examples that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, examples “comprising” or “having” an element or a plurality of elements having a particular condition can include additional elements not having that condition.

[0022] Examples of the present disclosure provide systems and methods that eliminate, minimize, or otherwise reduce human error in relation to visual inspection of parts. The systems and methods utilize digital cameras, which acquire data that is statistically interrogated and analyzed. The systems and methods described herein assist in the automation of visual inspection for surface and internal defect in parts (for example, translucent parts), and remove the subjective human element through the use of statistical analysis.

[0023] FIG. **1** illustrates a block diagram of a system **100**, according to an example of the present disclosure. The system **100** includes a light emitter **102** configured to emit light onto and/or into a part **106**. The light **104** reflects off, and/or passes through the part and is received by a detector **108**.

[0024] In at least one example, the light emitter **102** includes one or more light emitting diodes (LEDs) configured to emit the light. As another example, the light emitter **102** is or otherwise includes one more incandescent light bulbs. As another example, the light emitter **102** is or otherwise includes one or more lasers. As another example, the light emitter **102** is or otherwise includes one or more infrared red light sources.

[0025] The part **106** can be any part that is configured to be non-destructively inspected in relation to structural integrity. The part **106** is inspected to find any defects. The part **106** can be a composite structure, such as a composite panel, beam, or the like. The part **106** can be translucent or transparent, and configured to allow at least a portion of the light **104** to pass therethrough.

[0026] In at least one example, the light emitter **102** is separated from the part **106**. As another example, the light emitter **102** can be mounted on or within the part **106**.

[0027] In at least one example, the detector **108** is a camera, such as a digital photographic camera. As another example, the camera is a digital video camera.

[0028] The system **100** also includes a control unit **110** in communication with one or both of the light emitter **102** and/or the detector **108**, such as through one or more wired or wireless connections. In at least one example, the control unit **110** is also in communication with a user interface **112**, such as through one or more wired or wireless connections. The user interface **112** includes a display **114** and an input device **116**. In at least one example, the display **114** is an electronic device configured to electronically show images, videos, text, and/or the like. The display **114** can be a monitor, screen, television, touchscreen, and/or the like. The input device **116** can include a keyboard, mouse, stylus, touchscreen interface (that is, the input device **116** can be integral with the display **114**), and/or the like. The control unit **110** and the user interface **112** can be, or part of, a computer workstation. As another example, the control unit **110** and the user interface **112** can be, or part of, a handheld device, such as a smart phone, tablet, or the like.

[0029] In operation, the light emitter **102** emits the light onto and/or into the part **106**. The detector **108** acquires one or more digital images **118** of the illuminated part **106**. The control unit **110** receives the digital image(s) **118** from the detector **108**, and statistically analyzes the digital image(s) **118** to determine a structural integrity of the part **106**. For example, the control unit **110** analyzes the digital image(s) **118** to detect whether any defect exists in the part **106**.

[0030] In at least one example, the control unit **110** statistically validates the light emitter in relation to brightness and diffusion. In at least one example, the control unit **110** also analyzes the image(s) **118** in relation to a signal-to-noise ratio (SNR) to provide a quantitative measurement.

[0031] In at least one example, the control unit **110** analyzes the image(s) **118** in relation to a mean and/or a standard deviation. Further, the control unit **110** can compare the SNR against a background to verify there is a 3:1 SNR. The control unit **110** can also plot the mean and the standard deviation to provide a porosity curve.

[0032] As described herein, the system **100** includes the light emitter **102**, which is configured to emit light **104** onto or into the part **106** to illuminate the part **106**. The detector **108** is configured to acquire one or more digital images **118** of the part **106** as illuminated by the light **104**. The control unit **110** is in communication with the detector **108**. The control unit **110** is configured to receive the one or more digital images **118** of the part **106** as illuminated by the light **104** from the detector **108**. The control unit **110** is further configured to automatically analyze (without human intervention) the one or more digital images **118** of the part **106** as illuminated by the light **104** to detect one or more defects of the part **106**, and/or a porosity of the part **106**. In at least one example, the defects include a void, an inclusion, a delamination, a cut, a tear, or the like on an outer surface of the part **106**. In at least one example, the porosity relates to internal features (such as internal pores, openings, or the like) of the part **106**, such as portions between outer surfaces of the part **106**.

[0033] In at least one example, the control unit **110** is configured to show information regarding the part **106** on the display **114** of the user interface **112**. The information relates to the defects of the part **106**, and/or the porosity of the part **106**. In at least one example, the information includes statistical data, such as a mean, a standard deviation, a signal-to-noise ratio (SNR), and/or the like of one or more digital images **118** of the part **106**.

[0034] In at least one example, The system of claim **1**, the control unit **110** is can also subtract a digital image from a baseline image (such as stored in a memory) to remove known geometry effects to light transmission, and enhance defect detection.

[0035] FIG. **2** illustrates a flow chart of a qualitative assessment method, according to an example of the present disclosure. Referring to FIGS. **1** and **2**, before the control unit **110** receives the digital images **118** from the detector **108**, the system **100** is first qualitatively assessed. At **200**, the part **106** (such as a translucent composite part or a reference standard part) is positioned in relation to the light emitter **102**. For example, the part **106** is placed over the light emitter **102** at a known distance.

[0036] Next, at **202**, it is determined if the light **104** emitted by the light emitter **102** is of sufficient intensity and uniform throughout the part **106**. In at least one example, such a determination can be performed by an individual. As another example, the control unit **110** can automatically perform such step, such as by analyzing one or more test images of the part **106**, as received by the detector **108**. As an example, if the test images indicate a resulting predetermined light intensity (such as stored in memory), and the test images indicate a uniform light distribution throughout the part **106**, the control unit **110** determines that the light emitted by the light emitter **102** is of sufficient intensity and uniform through the part **106**.

[0037] If the control unit **110** compares the test images of the part **106** with the stored data regarding the light intensity and uniformity and determines that the light **104** emitted by the light emitter **102** is not of sufficient intensity and/or not uniform throughout the part **106**, the method proceeds from **202** to **204**, at which the light emitter **102** and/or the part **106** are repositioned in relation to one another (such as by raising or lowering the part **106** in relation to the light emitter **102**), and the method returns to **202**.

[0038] If, however, the control unit **110** determines that the light **104** is of sufficient intensity and uniform through the part **106** at **202**, the method proceeds from **202** to **206**, at which the control unit **110** determines if quality indicators within the test images are visible. The quality indicators can be features, such as spots, texts, markings, or the like of the part **106**. If the control unit **110** does not detect the quality indicators in the test images, the method proceeds from **206** to **208**, at which the light emitter **102** is changed to a different light emitter **102**, and the method proceeds to **204**.

[0039] If, however, the control unit **110** determines that the quality indicators are visible in the test image(s), the method then proceeds to a quantitative assessment.

[0040] FIG. **3** illustrates a flow chart of a quantitative assessment method, according to an example of the present disclosure. Referring to FIGS. **1-3**, if the quality indicators are visible in the test image(s) at **206**, the method proceeds to **210**, at which the control unit **110** receives one or more image digital image(s) **118** of the part **106**, as illuminated by the light **104** emitted by the light emitter **102**. The digital image(s) **118** are images of illuminated surfaces of the part **106**.

[0041] FIG. **4** illustrates a front view of a digital image **118**, according to an example of the present disclosure. Referring to FIGS. **1**, **3**, and **4**, at **212**, the control unit **110** selects a region of interest **119** in a digital image **118**. The region of interest **119** can be an area **121** bounded by a rectangle **123** of a predetermined size and shape. The control unit **110** automatically selects the region of interest **119**, such as based on a desired sample size representative of the acquired digital image **118**. In at least one example, the region of region of interest **119** is automatically selected as a limited sample area to provide a reference area that is devoid of extraneous features, such as wrinkles.

[0042] FIG. **5** illustrates a histogram **130**, according to an example of the present disclosure. Referring to FIGS. **1**, **3**, **4**, and **5**, at **214**, the control unit **110** generates a histogram **130** with respect to the digital image **118** and/or the region of interest **119**. The histogram **130** includes a Gaussian spread **132**, which is a light source variation within the region of interest **119**. The Gaussian spread typically increases if the light emitter **102** lacks diffusion grating, and/or if the distance from the light emitter **102** to the part **106** is improper. The histogram **130** includes a mean, and a standard deviation.

[0043] At **216**, the control unit **110** determines if the mean and the standard deviation of the histogram **130** are acceptable, as set forth in data stored in a memory. If the mean and the standard deviation of the histogram are not acceptable, the method returns to step **208** in FIG. **2**. If, however, the mean and the standard deviation are acceptable at **216**, the method proceeds to an inspection process, such as to determine the existence of a defect, such aa a void, delamination, or foreign material within the part **106** (as detected in the acquired digital image **118**), and/or porosity of the part **106** (as detected in the acquired digital image **118**).

[0044] FIG. **6** illustrates a flow chart of a method of detecting a defect in a digital image of a part, according to an example of the present disclosure. Referring to FIGS. **1**, **5**, and **6**, after confirming that the mean and the standard deviation of the histogram **130** is acceptable, at **218**, the control unit **110** converts the digital image(s) **118** of the part **106**, as illuminated by the light emitter **102**, to a gray scale. Next, the control unit **110** records the mean and the standard deviation of the region of interest **119** within the digital image **118**. At **222**, the control unit **110** then adjusts a threshold until the region of interest **119** reaches a predetermined red out, and then records the threshold for the mean value. Next, at **224**, the control unit **110** determines a signal to noise ratio (SNR) with respect to the region of interest **119**.

[0045] As an example, the SNR is a measure of the level of desired light intensity of the digital image(s) **118** with respect to light intensity of background light. The background light can be determined from data stored in a memory of a test image that does not include the part **106**.

[0046] At **226**, the control unit **110** then determines if the SNR is acceptable, based on predetermined stored data. For example, a 3:1 SNR is acceptable. If, at **226**, the SNR is not acceptable, the method proceeds to **228**, at which the light emitter **102** is re-verified (and the method can then return to step **200** in FIG. **2**).

[0047] If, however, the SNR is acceptable at **226**, the method proceeds to **230**, at which the part **106** is inspected, and digital images **118** are acquired at **232**. At **234**, the control unit **110** applies the threshold (as previously recorded) to the digital images **118** to evaluate discrepancies. Based on step **234**, the control unit **110** determines if the part **106** is acceptable (that is, free of defects) at **236**. If not, at **238**, the control unit **110** determines that the part **106** has failed (that is, includes one or more defects), and is held for engineering review. The control unit **110** may output data regarding such determination to the display **114**. For example, the control unit **110** may show one or more messages indicating such determination on the display **114**. If, however, the control unit **110** determines that the part is acceptable at **236**, at **240** the control unit **110** can pass the part **106** forward to a next production process. The control unit **110** may output data regarding such determination to the display **114**.

[0048] As described, the control unit **110** automatically analyzes the digital images **118** of the part **106**, as illuminated by the light **104** emitted from the light emitter **102**, to determine the existence of one or more defects (such as a void, delamination, foreign material, and/or the like) within the part **106**. In this manner, the control unit **110** eliminates, minimizes, or otherwise reduces human error in analyzing the part **106** for defects.

[0049] In addition to analyzing the digital image(s) **118** to determine the existence of defects, the control unit **110** can also automatically analyze the digital image(s) **118** to detect porosity of the part **106**. Optionally, the control unit **110** can automatically analyze the digital image(s) **118** to detect the porosity of the part **106** without determining the existence of defects of the part.

[0050] FIG. **7** illustrates a flow chart of a method of detecting a porosity of a part, according to an example of the present disclosure. Referring to FIGS. **1**, **5**, and **7**, after confirming that the mean and the standard deviation of the histogram **130** is acceptable, at **250**, the control unit **110** converts the digital image(s) **118** of the part **106**, as illuminated by the light emitter **102**, to a gray scale. At **252**, the control unit **110** records a mean value for each porosity coupon. At **252**, the control unit **110** also records a standard deviation for a **0**% porosity coupon. Then, at **254**, the control unit **110** generates new (or validates current) porosity curves.

[0051] Porosity coupons contain fine porosity pores at known values from **0**% to a maximum which results in decreasing light transmission. These reduced transmission values are used to create and calibrate light transmission or absorption (attenuation) curves.

[0052] In at least one a porosity coupon is a test portion of the part **106**. In at least one example, the porosity coupon can be a region of the part **106** within the region of interest **119** of the digital image **118** (shown in FIG. **4**). The region of interest **119** can be the porosity coupon. The mean decreases, and the standard deviation increases, as porosity increases.

[0053] Next, at **256**, the control unit **110** determines a signal to noise ratio (SNR) with respect to the region of interest **119** and/or the porosity coupon. At **258**, the control unit **110** then determines if the SNR is acceptable, based on predetermined stored data. For example, a 3:1 SNR is acceptable. If, at **258**, the SNR is not acceptable, the method proceeds to **260**, at which the light emitter **102** is re-verified (and the method can then return to step **200** in FIG. **2**).

[0054] If, however, the SNR is acceptable at **258**, the method proceeds to **262**, at which the part **106** is inspected, and digital images **118** are acquired. At **266**, the control unit **110** applies the threshold (as previously recorded) to the digital images **118** to evaluate discrepancies. Based on step **266**, the control unit **110** determines if the part **106** is acceptable (that is, free of defects) at **268**. If not, at **270**, the control unit **110** determines that the part **106** has failed (that is, includes undesirable porosity), and is held for engineering review. The control unit **110** may output data regarding such determination to the display **114**. For example, the control unit **110** may show one or more messages indicating such determination on the display **114**. If, however, the control unit **110** determines that the part is acceptable at **236**, at **272**, the control unit **110** can pass the part **106** forward to a next production process. The control unit **110** may output data regarding such determination to the display **114**.

[0055] Referring to FIGS. **1-7**, examples of the present disclosure provide quantitative methods to conduct non-destructive inspection (NDI) of parts **106** (such as translucent composite parts) using statistical analysis. The systems and methods described herein remove subjective variability inherent in manual visual inspection of parts.

[0056] The systems and methods described herein include the control unit **110**, which is configured to automatically inspect the part **106**, such as through statistical analysis of one or more digital images **118** of the part **106**. The control unit **110** analyzes the digital images **118** to detect surface and/or internal defects of the part **106**. The control unit **110** analyzes the digital image(s) **118** to determine a mean and a standard deviation of the digital image(s) **118**. In at least one example, the control unit **110** determines the SNR, and compares the SNR against a background to verify that the SNR is acceptable. In at least one example, the control unit **110** also plots the mean and the standard deviation to create a porosity curve.

[0057] FIG. **8** illustrates a schematic block diagram of the control unit **110**, according to an example of the present disclosure. In at least one example, the control unit **110** includes at least one processor **300** in communication with a memory **302**. The memory **302** stores instructions **304**, received data **306**, and generated data **308**. The control unit **110** shown in FIG. **8** is merely exemplary, and non-limiting.

[0058] As used herein, the term “control unit,” “central processing unit,” “CPU,” “computer,” or the like may include any processor-based or microprocessor-based system including systems using microcontrollers, reduced instruction set computers (RISC), application specific integrated circuits (ASICs), logic circuits, and any other circuit or processor including hardware, software, or a combination thereof capable of executing the functions described herein. Such are exemplary only, and are thus not intended to limit in any way the definition and/or meaning of such terms. For example, the control unit **110** may be or include one or more processors that are configured to control operation, as described herein.

[0059] The control unit **110** is configured to execute a set of instructions that are stored in one or more data storage units or elements (such as one or more memories), in order to process data. For example, the control unit **110** may include or be coupled to one or more memories. The data storage units may also store data or other information as desired or needed. The data storage units may be in the form of an information source or a physical memory element within a processing machine.

[0060] The set of instructions may include various commands that instruct the control unit **110** as a processing machine to perform specific operations such as the methods and processes of the various examples of the subject matter described herein. The set of instructions may be in the form of a software program. The software may be in various forms such as system software or application software. Further, the software may be in the form of a collection of separate programs, a program subset within a larger program, or a portion of a program. The software may also include modular programming in the form of object-oriented programming. The processing of input data by the processing machine may be in response to user commands, or in response to results of previous processing, or in response to a request made by another processing machine.

[0061] The diagrams of examples herein may illustrate one or more control or processing units, such as the control unit **110**. It is to be understood that the processing or control units may represent circuits, circuitry, or portions thereof that may be implemented as hardware with associated instructions (e.g., software stored on a tangible and non-transitory computer readable storage medium, such as a computer hard drive, ROM, RAM, or the like) that perform the operations described herein. The hardware may include state machine circuitry hardwired to perform the functions described herein. Optionally, the hardware may include electronic circuits that include and/or are connected to one or more logic-based devices, such as microprocessors, processors, controllers, or the like. Optionally, the control unit **110** may represent processing circuitry such as one or more of a field programmable gate array (FPGA), application specific integrated circuit (ASIC), microprocessor(s), and/or the like. The circuits in various examples may be configured to execute one or more algorithms to perform functions described herein. The one or more algorithms may include aspects of examples disclosed herein, whether or not expressly identified in a flowchart or a method.

[0062] As used herein, the terms “software” and “firmware” are interchangeable, and include any computer program stored in a data storage unit (for example, one or more memories) for execution by a computer, including RAM memory, ROM memory, EPROM memory, EEPROM memory, and non-volatile RAM (NVRAM) memory. The above data storage unit types are exemplary only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0063] In at least one example, components of the system **100**, such as the control unit **110**, provide and/or enable a computer system to operate as a special computer system for determining a structural integrity of the part **106**, such as through analysis of the digital image(s) **118** of the part **106**, as illuminated by the light **104** emitted by the light emitter **102**. The control unit **110** improves upon standard computing devices by determining such information and automatically communicating with individuals in an efficient and effective manner.

[0064] In at least one example, all or part of the systems and methods described herein are or otherwise include an artificial intelligence (AI) or machine-learning system that can automatically perform the operations of the methods also described herein. In at least one example, the control unit **110** can be or otherwise include a deterministic or rules based evaluation system. In at least one example, the control unit **110** can be an artificial intelligence or machine learning system. These types of systems may be trained from outside information and/or self-trained to repeatedly improve the accuracy with how data is analyzed to determine and present the relevant information to users. For example, an AI control unit **110** can be trained to automatically detect regions of interest within images, statistical features of the images, and the like. Over time, these systems can improve by determining and communicating with increasing accuracy and speed, thereby significantly reducing the likelihood of any potential errors. For example, the AI or machine-learning systems can learn and determine models, associate such models with received data, and determine potential conflicts. The AI or machine-learning systems described herein may include technologies enabled by adaptive predictive power and that exhibit at least some degree of autonomous learning to automate and/or enhance pattern detection (for example, recognizing irregularities or regularities in data), customization (for example, generating or modifying rules to optimize record matching), and/or the like. The systems may be trained and re-trained using feedback from one or more prior analyses of the data, ensemble data, and/or other such data. Based on this feedback, the systems may be trained by adjusting one or more parameters, weights, rules, criteria, or the like, used in the analysis of the same. This process can be performed using the data and ensemble data instead of training data, and may be repeated many times to repeatedly improve the determinations and communications described herein. The training minimizes conflicts and interference by performing an iterative training algorithm, in which the systems are retrained with an updated set of data, and based on the feedback examined prior to the most recent training of the systems. This provides a robust analysis model that can better determine and present statistical information regarding the digital image(s) **118**.

[0065] Further, the disclosure comprises examples according to the following clauses: [0066] Clause 1. A system comprising:

a light emitter configured to emit light onto or into a part to illuminate the part;

a detector configured to acquire one or more digital images of the part as illuminated by the light; and

a control unit in communication with the detector, wherein the control unit is configured to: [0067] receive the one or more digital images of the part as illuminated by the light from the detector, and [0068] automatically analyze the one or more digital images of the part as illuminated by the light to one or both of detect one or more defects of the part, or determine a porosity of the part. [0069] Clause 2. The system of Clause 1, wherein the control unit is configured to detect one or more defects of the part and determine the porosity of the part. [0070] Clause 3. The system of Clauses 1 or 2, wherein the detector comprises a digital camera. [0071] Clause 4. The system of any of Clauses 1-3, wherein the light emitter is separated from the part. [0072] Clause 5. The system of any of Clauses 1-4, further comprising a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the part on the display. [0073] Clause 6. The system of any of Clauses 1-5, wherein the control unit is configured to automatically analyze the one or more digital images by determining a mean and a standard deviation of one or more portions of the one or more digital images. [0074] Clause 7. The system of any of Clauses 1-6, wherein the control unit is configured to automatically analyze the one or more digital images by determining a signal-to-noise ratio (SNR) of one or more portions of the one of more digital images. [0075] Clause 8. The system of any of Clauses 1-7, wherein the control unit is configured to analyze the one or more digital images to generate a histogram in relation to one or more portions of the one or more digital images, and determine a mean and a standard deviation from the histogram. [0076] Clause 9. The system of any of Clauses 1-8, wherein the control unit is an artificial intelligence or machine learning system. [0077] Clause 10. A method for a system comprising:

a light emitter configured to emit light onto or into a part to illuminate the part;

a detector configured to acquire one or more digital images of the part as illuminated by the light; and

a control unit in communication with the detector, wherein the control unit is configured to: [0078] receive the one or more digital images of the part as illuminated by the light from the detector, and [0079] automatically analyze the one or more digital images of the part as illuminated by the light to one or both of detect one or more defects of the part, or determine a porosity of the part,

the method comprising:

emitting, by the light emitter, the light onto or into the part to illuminate the part;

acquiring, by the detector, the one or more digital images of the part as illuminated by the light;

receiving, by the control unit, the one or more digital images of the part as illuminated by the light from the detector; and

automatically analyzing, by the control unit, the one or more digital images of the part as illuminated by the light, wherein said automatically analyzing comprises one or both of detecting the one or more defects of the part, or determining the porosity of the part. [0080] Clause 11. The method of Clause 10, wherein said automatically analyzing comprises the detecting and the determining. [0081] Clause 12. The method of Clauses 10 or 11, wherein the detector comprises a digital camera. [0082] Clause 13. The method of any of Clauses 10-12, wherein the light emitter is separated from the part. [0083] Clause 14. The method of any of Clauses 10-13, further comprising showing, by the control unit, information regarding the part on a display of a user interface in communication with the control unit. [0084] Clause 15. The method of any of Clauses 10-14, further comprising determining a mean and a standard deviation of one or more portions of the one or more digital images. [0085] Clause 16. The method of any of Clauses 10-15, wherein said automatically analyzing comprises determining a signal-to-noise ratio (SNR) of one or more portions of the one of more digital images. [0086] Clause 17. The method of any of Clauses 10-16, wherein said automatically analyzing comprises:

[0087] generating a histogram in relation to one or more portions of the one or more digital images; and determining a mean and a standard deviation from the histogram. [0088] Clause 18. The method of any of Clauses 10-17, wherein the control unit is an artificial intelligence or machine learning system. [0089] Clause 19. A system comprising:

a light emitter configured to emit light onto or into a part to illuminate the part, wherein the light emitter is separated from the part;

a detector configured to acquire one or more digital images of the part as illuminated by the light, wherein the detector comprises a digital camera;

a control unit in communication with the detector, wherein the control unit is configured to: [0090] receive the one or more digital images of the part as illuminated by the light from the detector, and [0091] automatically analyze the one or more digital images of the part as illuminated by the light to detect one or more defects of the part, and determine a porosity of the part, wherein the control unit is configured to automatically analyze the one or more digital images by determining a mean and a standard deviation of one or more portions of the one or more digital images, and a signal-to-noise ratio (SNR) of one or more portions of the one of more digital images; and

a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the part on the display. [0092] Clause 20. The system of Clause 19, wherein the control unit is configured to analyze the one or more digital images to generate a histogram in relation to one or more portions of the one or more digital images, and determine the mean and the standard deviation from the histogram. [0093] Clause 21. The system or method of any of the Clauses, wherein the control unit is further configured to subtract the one or more digital images from a baseline image to remove known geometry effects to light transmission, and enhance defect detection.

[0094] As described herein, examples of the present disclosure provide improved systems and methods for inspecting component parts to ensure structural integrity. Further, examples of the present disclosure provide efficient and effective systems and methods for inspecting component parts.

[0095] While various spatial and directional terms, such as top, bottom, lower, mid, lateral, horizontal, vertical, front and the like can be used to describe examples of the present disclosure, it is understood that such terms are merely used with respect to the orientations shown in the drawings. The orientations can be inverted, rotated, or otherwise changed, such that an upper portion is a lower portion, and vice versa, horizontal becomes vertical, and the like.

[0096] As used herein, a structure, limitation, or element that is “configured to” perform a task or operation is particularly structurally formed, constructed, or adapted in a manner corresponding to the task or operation. For purposes of clarity and the avoidance of doubt, an object that is merely capable of being modified to perform the task or operation is not “configured to” perform the task or operation as used herein.

[0097] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described examples (and/or aspects thereof) can be used in

combination with each other. In addition, many modifications can be made to adapt a particular situation or material to the teachings of the various examples of the disclosure without departing from their scope. While the dimensions and types of materials described herein are intended to define the aspects of the various examples of the disclosure, the examples are by no means limiting and are exemplary examples. Many other examples will be apparent to those of skill in the art upon reviewing the above description. The scope of the various examples of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims and the detailed description herein, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.
[0098] This written description uses examples to disclose the various examples of the disclosure, including the best mode, and also to enable any person skilled in the art to practice the various examples of the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the various examples of the disclosure is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if the examples have structural elements that do not differ from the literal language of the claims, or if the examples include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A system comprising: a light emitter configured to emit light onto or into a part to illuminate the part; a detector configured to acquire one or more digital images of the part as illuminated by the light; and a control unit in communication with the detector, wherein the control unit is configured to: receive the one or more digital images of the part as illuminated by the light from the detector, and automatically analyze the one or more digital images of the part as illuminated by the light to one or both of detect one or more defects of the part, or determine a porosity of the part.
2. The system of claim 1, wherein the control unit is further configured to subtract the one or more digital images from a baseline image to remove known geometry effects to light transmission and enhance defect detection.
3. The system of claim 1, wherein the control unit is configured to detect one or more defects of the part and determine the porosity of the part.
4. The system of claim 1, wherein the detector comprises a digital camera.
5. The system of claim 1, wherein the light emitter is separated from the part.
6. The system of claim 1, further comprising a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the part on the display.
7. The system of claim 1, wherein the control unit is configured to automatically analyze the one or more digital images by determining a mean and a standard deviation of one or more portions of the one or more digital images.
8. The system of claim 1, wherein the control unit is configured to automatically analyze the one or more digital images by determining a signal-to-noise ratio (SNR) of one or more portions of the one of more digital images.
9. The system of claim 1, wherein the control unit is configured to analyze the one or more digital images to generate a histogram in relation to one or more portions of the one or more digital

images, and determine a mean and a standard deviation from the histogram.
10. The system of claim 1, wherein the control unit is an artificial intelligence or machine learning system.
11. A method for a system comprising: a light emitter configured to emit light onto or into a part to illuminate the part; a detector configured to acquire one or more digital images of the part as illuminated by the light; and a control unit in communication with the detector, wherein the control unit is configured to: receive the one or more digital images of the part as illuminated by the light from the detector, and automatically analyze the one or more digital images of the part as illuminated by the light to one or both of detect one or more defects of the part, or determine a porosity of the part, the method comprising: emitting, by the light emitter, the light onto or into the part to illuminate the part; acquiring, by the detector, the one or more digital images of the part as illuminated by the light; receiving, by the control unit, the one or more digital images of the part as illuminated by the light from the detector; and automatically analyzing, by the control unit, the one or more digital images of the part as illuminated by the light, wherein said automatically analyzing comprises one or both of detecting the one or more defects of the part, or determining the porosity of the part.
12. The method of claim 11, wherein said automatically analyzing comprises the detecting and the determining.
13. The method of claim 11, wherein the detector comprises a digital camera.
14. The method of claim 11, wherein the light emitter is separated from the part.
15. The method of claim 11, further comprising showing, by the control unit, information regarding the part on a display of a user interface in communication with the control unit.
16. The method of claim 11, wherein said automatically analyzing comprises determining a mean and a standard deviation of one or more portions of the one or more digital images.
17. The method of claim 11, wherein said automatically analyzing comprises determining a signal-to-noise ratio (SNR) of one or more portions of the one of more digital images.
18. The method of claim 11, wherein said automatically analyzing comprises: generating a histogram in relation to one or more portions of the one or more digital images; and determining a mean and a standard deviation from the histogram.
19. The method of claim 11, wherein the control unit is an artificial intelligence or machine learning system.
20. A system comprising: a light emitter configured to emit light onto or into a part to illuminate the part, wherein the light emitter is separated from the part; a detector configured to acquire one or more digital images of the part as illuminated by the light, wherein the detector comprises a digital camera; a control unit in communication with the detector, wherein the control unit is configured to: receive the one or more digital images of the part as illuminated by the light from the detector, and automatically analyze the one or more digital images of the part as illuminated by the light to detect one or more defects of the part, and determine a porosity of the part, wherein the control unit is configured to automatically analyze the one or more digital images by determining a mean and a standard deviation of one or more portions of the one or more digital images, and a signal-to-noise ratio (SNR) of one or more portions of the one of more digital images; and a user interface in communication with the control unit, wherein the user interface comprises a display, and wherein the control unit is configured to show information regarding the part on the display.
21. The system of claim 20, wherein the control unit is configured to analyze the one or more digital images to generate a histogram in relation to one or more portions of the one or more digital images, and determine the mean and the standard deviation from the histogram.